

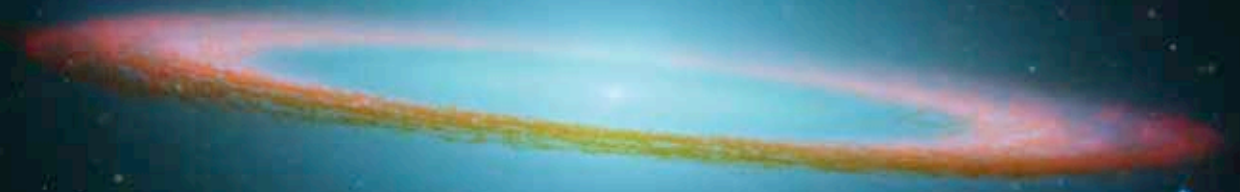
Cooler Master adds more coolers

Cooler Master added two new coolers namely the 240M and 120XL to its Seidon line of all-in-one liquid coolers
<http://dgit.in/WUeE8C>

Vine to be launched by Twitter

Users on Twitter will soon be able to share video clips on their network and other websites using a new Video app called Vine <http://dgit.in/W5z5RA>

Gazing into the void



The complexities involved in the construction of a telescope go far beyond mere optics and glass. We take a look at some of the challenges faced by engineers to this day

Anirudh Regidi

anirudhregidi@thinkdgit.com

Achieving feasible space travel is a dream that is still very far from reality, but we can still gaze into the void of the cosmos and dream of one day reaching those distant stars. The only way we can do that now is by the use of telescopes. These have evolved from humble, hand-held devices with apertures of a few mm to the radio behemoths of today with apertures that are many times the diameter of the Earth.

Light

To understand how a telescope works we would need a better understanding of light in the context of space. What we perceive as light and colour is electromagnetic radiation within a particular frequency (or wavelength) that our eyes respond to. X-Rays, radio waves, infra-red, etc. are all the same thing. Electro-magnetic radiation, only, they're travelling at a different wavelength. To put it another way, our eyes are electro-magnetic sensors that are sensitive to a particular range of wavelengths, the same way

that your cell phone's FM radio's antenna can pick up radio waves in the correct band.

The sky at night is lovely to behold, if you can see the stars through the thick layer of smog that is, but the one stand out feature of the night sky is that it is dark. Why is that so? If the arc of a grain of sand held at arm's length can contain thousands of galaxies, why do we not see them all when we look up? The answer is surprisingly simple. Almost all the stars in space are moving away from us at very high-speeds, speeds that are unimaginable for us on Earth and easily in the range of hundreds of thousands of kilometres an hour and the farther the star, the faster it is moving away from us.

This changing speed causes a change in the wavelength of light where it is "red-shifted" (due to the Doppler Effect). This "red-shift" causes the electro-magnetic spectrum to shift from the visible range, the range our eyes are sensitive to, to infra-red to the microwave and even radio wavelength. Of course, the exact opposite happens to objects that are coming closer to us and the radiation from those objects shift from the visible spectrum to UV, X-ray, and gamma ray and beyond. There are more factors that

determine the nature of this radiation, but to enumerate them would be way beyond the scope of this article. Suffice to say that since our eyes are only receptive of a fraction of the electromagnetic spectrum, we only see a fraction of the actual "light" in space.

Types of telescopes

Based on the above understanding of how light and electro-magnetic radiation work, telescopes have to be specifically designed to receive and interpret radiation in specific bands. As such, telescopes are classified as:

- Optical
- Radio
- X-Ray
- Gamma-Ray
- Sub-mm and so on.

Of note are the radio telescope arrays that are massive structures that can cover hundreds of square kilometres. These telescopes are essentially, large antennae that are separated by many kilometres, with some even positioned in space. These antennae pick up incoming radio signals and can, mathematically speaking, produce apertures that are larger than the diameter of the Earth. Managing